Quality Information Document

Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment Monitoring Service

Sea Surface Temperature OMI for the EuropeanNorthwest Shelf

OMI_CLIMATE_SST_NORTHWESTSHELF_area_averaged_anomalies

OMI_CLIMATE_SST_NORTHWESTSHELF_trend

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
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I INTRODUCTION

I.1 Product name

OMI_CLIMATE_SST_NORTHWESTSHELF_ area_averaged_anomalies

OMI_CLIMATE_SST_NORTHWESTSHELF_ trend

I.2 Indicator evaluation method

The European Northwest Shelf Sea Surface Temperature (SST) Ocean Monitoring Indicators (OMIs) are derived from the CMEMS Reprocessed European North West Shelf Iberia-Biscay-Irish areas (REP ATL) SST L4 product[1]. The REP ATL L4 product provides the SSTs that are used to compute all the IBI SST OMIs, i.e. the time series of SST anomalies (nws_omi_tempsal_sst_area_averaged_anomalies) and the 2D map of SST trend (nws_omi_tempsal_sst_trend). Anomalies are computed against the 1991-2020 reference period. In detail, daily maps of REP ATL L4 data from 1982 were averaged to obtain SST monthly maps,which were then used to build the time series of monthly mean SST values averaged over the entire basin.

Trend analysis has been performed by using the X-11 seasonal adjustment procedure (Pezzulli et al. 2005). This procedure decomposes the input SST time series into the sum of three distinct components:

$$SSTt = Tt + St + It,$$

where Tt is the trend component, St and It the seasonal and the irregular components. The X-11 procedure has the effect of filtering the input time series with Tt acting as a low bandpass filter for interannual variations. The only difference with respect to the standard X-11 procedure is that we substituted the 13-term moving-average window (used to estimate Tt) with a 25-term moving-average window, i.e. from 1-year to 2-year low-pass filter. This makes more robust the estimation of the trend component Tt.

[1] cmems_SST_ATL_SST_L4_REP_OBSERVATIONS_010_026

Mann-Kendall test (Mann, 1945; Kendall, 1975) was applied to Tt in order to assess whether there is a monotonic upward or downward trend over time and Sens's method (Sen, 1968) was subsequently applied to estimate the slope (°C/year) of the trend component Tt and its 95% confidence interval.

I.3 CMEMS products used as input for the computations

Product name (and Type)	Documentation
cmems_SST_ATL_SST_L4_REP_OBSERVATIONS_010_026	QUID: https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-ATL-QUID-010-026.pdf PUM: https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-ATL-PUM-010-026.pdf

II SCIENTIFIC VALIDATION

The reliability of the NORTHWESTSHELF SST OMI products is strictly related to the quality and accuracy of the REP ATL L4 SST product, which is validated in the CMEMS REP ATL QUID (see http://marine.copernicus.eu/documents/QUID/CMEMS-SST-QUID-010-026.pdf). The accuracy of the REP ATL L4 product has been assessed by comparison to independent in situ data for the period 2005-2016. Results show an overall mean bias and root mean square difference (RMSD) of the differences between REP ATL and in situ data of -0.07 and 0.46 K, respectively. The time series of monthly averaged mean bias and RMSD is shown in Figure 1. The overall validation indicates that this product represents a valuable SST dataset for climate studies.

We recall that the REP ATL L4 product is a collection of daily (night-time) interpolated (level-4, L4) satellite-based estimates of the foundation SST over the European North West Shelf Iberia-Biscay-Irish areas (ATL) at 0.05° grid covering the period 1982-2023. This product is built from a consistent reprocessing of the level-3 (merged multi-sensor, L3) climate data record provided by the ESA Climate Change Initiative (CCI) (Merchant et al., 2019) and Copernicus Climate Change Service (C3S) initiatives.

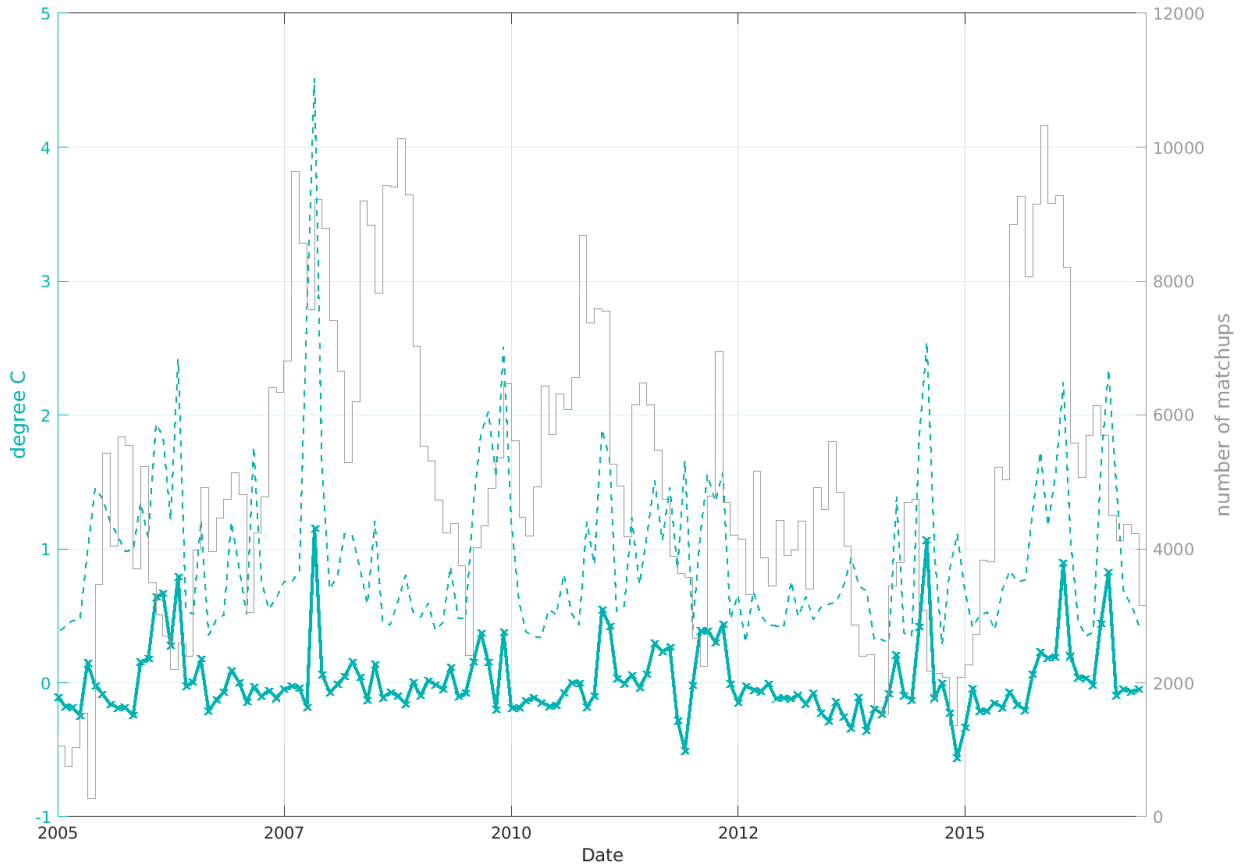

Figure 1: Monthly RMSD (dashed blue) and mean bias (blue line) of the differences (°C) between the REP ATL L4 and in situ data over the period 2005-2016. The number of matchups used to compute the statistics is indicated by the gray line.

Product used for indicator evaluation:

The NORTHWESTSHELF SST OMI products are derived from the CMEMS REP ATL L4 product, which is a stable and consistent long-term SST dataset covering the European North West Shelf Iberai-Biscay-Irish areasand developed for climate applications. This REP ATL L4 product provides daily, nighttime, 5 km resolution reprocessed gap-free (L4) SST maps over the European North West Shelf Iberia-Biscay-Irish Seas. A detailed description of this product and its assessment can be found in the CMEMS PUM and QUID (http://marine.copernicus.eu/documents/PUM/CMEMS-SST-PUM-010-026.pdf; http://marine.copernicus.eu/documents/QUID/CMEMS-SST-QUID-010-026.pdf.

III REFERENCES

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